Table 14: **Landslide Hazard Mitigation Measures (adapted from Popescu & Sasahara, 2009)**

Physical process	Brief description
Surface protection; control of surface erosion	• Vegetation (hydroseeding, turfing, trees/bushes) • Fascines/brush. • Geosynthetics. • Substitution; drainage blanket • beach replenishment; rip-rap. • Dentition
Modifying the geometry and/or mass distribution	• Removal of material from the area driving the landslide (with possible substitution by lightweight fill). • Addition of material to the area maintaining stability, with or without gravity, catilever, crib/cellular and/or reinforced soil walls. • Reduction of the general slope angle. • Scaling (removal of loose/unstable blocks/boulders).
Modifying surface water regime – surface drainage	• Diversion channels • Check dams • Surface drains (ditches, piping) to divert water from flowing onto the slide area. • Sealing tension cracks. • Impermeabilization. (*) • Vegetation. (*) Note (*): associated with control of surface erosion
Modifying groundwater regime – deep drainage	• Shallow or deep trenches filled with coarse grained free-draining geomaterials and geosynthetics • Subhorizontal drains • Vertical small diameter wells; self draining (where they provide relief to artesian pressures or underdrainage to a perched acquifer) or drained by siphoning, electropneumatic or electromechanical pumps • Vertical medium diameter wells with gravity drainage through a base collector • Caissons (large diameter wells), with or without secondary subhorizontal drains and gravity drainage • Drainage tunnels, galleries, adits, with or without secondary subhorizontal or subvertical drains and/or as gravity outlet for wells drilled from the surface
Modifying the mechanical characteristics of the unstable mass	• Substitution • Compaction • Deep mixing with lime and/or cement • Permeation or pressure grouting with cementitiuous or chemical binders • Jet grouting • Modification of the groundwater chemistry
Transfer of loads to more competent strata	• Shear keys: counterforts, piles; barrettes (diaphragm walls); caissons • Anchors: soil nails; dowels, rock bolts; multistrand anchors (with or without facing consisting of plates, nets, reinforced shotcrete) • Anchored walls (combination of anchors and shear keys)